

19 datasets · 1,707 sequences

Laparoscopy

rigid MIS
CholecT50 / EndoVis

GI endoscopy

flexible / capsule
Kvasir / C3VD

Bronchoscopy

ION / airway
(private)

Shared encoder

V-JEPA 2 ViT-L
frozen, $D = 1024$
dense tokens z

Causal residual L1

short-horizon forecast
GPT-style rollout

Coarse L2

mid-horizon anatomy

SE(3) dynamics

block-causal ensemble
risk + covariance

Capability 1

forecast scene evolution
 $0.978 \cos, p < 10^{-80}$

Capabilities 2 + 3

action evaluation 83.1%
offline navigation 51.5%

Loss-of-view warning

anticipate before losing sight

Camera-handling training

score a motion's visual effect

Navigation assistance

model-based planning prototype

Input-sensitivity tests

history / action / domain

Reprojection pose gate

$10.2 \text{ px} \rightarrow 0.21 \text{ px}$

Matched negative banks

fixed, same-sequence

Grouped CV + frozen test

no test-set tuning

Passive-video forecast (validated)

Physical grounding (audited; pose/depth-gated)

Correctness-audit gates (Section: Correctness auditing and evaluation gates)

Clinical application pathways